SQL FUNCTIONS: 25 PRACTICALS

MySQL command-line code with sample output

Instructions: Use MySQL 8.0+ for these examples. Run the setup once, then execute each practical. Outputs are representative and may vary slightly by MySQL version or your existing data.

Setup: database, tables and sample records

```
CREATE DATABASE IF NOT EXISTS CollegeDB;
USE CollegeDB;

CREATE TABLE IF NOT EXISTS Student (
  Student_ID INT PRIMARY KEY,
  First_Name VARCHAR(40),
  Last_Name VARCHAR(40),
  Name VARCHAR(80),
  Age INT,
  Course VARCHAR(40),
  City VARCHAR(40)
);

CREATE TABLE IF NOT EXISTS Employee (
  Emp_ID INT PRIMARY KEY,
  Emp_Name VARCHAR(80),
  Department VARCHAR(40),
  Salary DECIMAL(10,2),
  Join_Date DATE
);

INSERT INTO Student VALUES
(1,'Amit','Sharma','Amit Sharma',20,'BCA','Meerut'),
(2,'Neha','Verma','Neha Verma',21,'B.Tech','Delhi'),
(3,'Rahul','Singh','Rahul Singh',22,'BCA','Jaipur'),
(4,'Priya','Gupta','Priya Gupta',20,'BBA','Meerut'),
(5,'Karan','Mehta','Karan Mehta',23,'B.Tech','Delhi');

INSERT INTO Employee VALUES
(101,'Rohit Kumar','IT',45678.65,'2022-04-15'),
(102,'Pooja Singh','HR',38250.40,'2021-09-01'),
(103,'Aman Verma','Sales',52700.75,'2020-01-20'),
(104,'Sneha Gupta','IT',61500.00,'2019-07-10');
```

If the tables already contain these IDs, skip the sample INSERT statements or use a fresh practice database.

A. Number Functions

1. Demonstrate ABS()

```
SELECT ABS(-25) AS Absolute_Value;
```

Sample output:

```
Absolute_Value
25
```

2. Demonstrate ROUND()

```
SELECT ROUND(123.456, 2) AS Rounded_Value;
```

Sample output:

```
Rounded_Value
123.46
```

3. Demonstrate CEIL() and FLOOR()

```
SELECT CEIL(12.3) AS Ceiling_Value, FLOOR(12.9) AS Floor_Value;
```

Sample output:

```
Ceiling_Value | Floor_Value
13            | 12
```

4. Demonstrate MOD()

```
SELECT MOD(17, 5) AS Remainder;
```

Sample output:

```
Remainder
2
```

5. Calculate rounded salary of employees

```
SELECT Emp_ID, Emp_Name, ROUND(Salary) AS Rounded_Salary FROM Employee;
```

Sample output:

```
101 | Rohit Kumar | 45679
102 | Pooja Singh | 38250
103 | Aman Verma  | 52701
104 | Sneha Gupta | 61500
```

B. Aggregate Functions

6. Count total students

```
SELECT COUNT(*) AS Total_Students FROM Student;
```

Sample output:

```
Total_Students
5
```

7. Sum of all employee salaries

```
SELECT SUM(Salary) AS Total_Salary FROM Employee;
```

Sample output:

```
Total_Salary
198129.80
```

8. Average employee salary

```
SELECT ROUND(AVG(Salary), 2) AS Average_Salary FROM Employee;
```

Sample output:

```
Average_Salary
49532.45
```

9. Highest employee salary

```
SELECT MAX(Salary) AS Highest_Salary FROM Employee;
```

Sample output:

```
Highest_Salary
61500.00
```

10. Lowest employee salary

```
SELECT MIN(Salary) AS Lowest_Salary FROM Employee;
```

Sample output:

```
Lowest_Salary
38250.40
```

C. Character Functions

11. UPPER() on student names

```
SELECT Name, UPPER(Name) AS Upper_Name FROM Student;
```

Sample output:

```
Amit Sharma | AMIT SHARMA
Neha Verma  | NEHA VERMA
Rahul Singh | RAHUL SINGH
Priya Gupta | PRIYA GUPTA
Karan Mehta | KARAN MEHTA
```

12. LOWER() on student names

```
SELECT Name, LOWER(Name) AS Lower_Name FROM Student;
```

Sample output:

```
Amit Sharma | amit sharma
Neha Verma  | neha verma
Rahul Singh | rahul singh
Priya Gupta | priya gupta
Karan Mehta | karan mehta
```

13. LENGTH() on student names

```
SELECT Name, LENGTH(Name) AS Name_Length FROM Student;
```

Sample output:

```
Amit Sharma | 11
Neha Verma  | 10
Rahul Singh | 11
Priya Gupta | 11
Karan Mehta | 10
```

14. CONCAT() using first and last names

```
SELECT CONCAT(First_Name, ' ', Last_Name) AS Full_Name FROM Student;
```

Sample output:

```
Full_Name
Amit Sharma
Neha Verma
Rahul Singh
Priya Gupta
Karan Mehta
```

15. SUBSTRING()/SUBSTR()

```
SELECT Name, SUBSTRING(Name, 1, 4) AS First_Four FROM Student;
```

Sample output:

```
Amit Sharma | Amit
Neha Verma  | Neha
Rahul Singh | Rahu
Priya Gupta | Priy
Karan Mehta | Kara
```

D. Conversion Functions

16. Convert a number to a string

```
SELECT CAST(12345 AS CHAR) AS Number_As_Text;
```

Sample output:

```
Number_As_Text
12345
```

17. Convert a string to a number

```
SELECT CAST('250' AS UNSIGNED) + 50 AS Numeric_Result;
```

Sample output:

```
Numeric_Result
300
```

18. Convert a date into a different format

```
SELECT DATE_FORMAT('2024-03-15', '%d/%m/%Y') AS Formatted_Date;
```

Sample output:

```
Formatted_Date
15/03/2024
```

19. Demonstrate CAST()

```
SELECT CAST('123.45' AS DECIMAL(8,2)) AS Converted_Number;
```

Sample output:

```
Converted_Number
123.45
```

20. Demonstrate CONVERT()

```
SELECT CONVERT('2024-12-31', DATE) AS Converted_Date;
```

Sample output:

```
Converted_Date
2024-12-31
```

E. Date Functions

21. Display current date

```
SELECT CURDATE() AS Current_Date;
```

Sample output:

```
Current_Date
(Displays the date on your MySQL server)
```

22. Extract year, month and day

```
SELECT YEAR('2024-03-15') AS Year, MONTH('2024-03-15') AS Month, DAY('2024-03-15') AS Day;
```

Sample output:

```
Year | Month | Day
2024 | 3     | 15
```

23. Difference between two dates

```
SELECT DATEDIFF('2024-04-10', '2024-04-01') AS Days_Difference;
```

Sample output:

```
Days_Difference
9
```

24. Add days to a date

```
SELECT DATE_ADD('2024-03-15', INTERVAL 10 DAY) AS New_Date;
```

Sample output:

```
New_Date
2024-03-25
```

25. Display joining dates in required format

```
SELECT Emp_Name, DATE_FORMAT(Join_Date, '%d-%m-%Y') AS Joining_Date FROM Employee;
```

Sample output:

```
Rohit Kumar | 15-04-2022
Pooja Singh | 01-09-2021
Aman Verma  | 20-01-2020
Sneha Gupta | 10-07-2019
```

Important notes

• The sample employee salaries total 198129.80; average salary is 49532.45.

• In MySQL, LENGTH() counts bytes; for these English names it matches the character count.

• DATE_FORMAT(), CURDATE(), YEAR(), MONTH(), DAY(), DATEDIFF(), and DATE_ADD() are MySQL functions.

• Re-running the sample INSERT statements can cause duplicate primary-key errors; use a fresh database or skip those inserts.